

Detection of Faults in an Induction Motor using Signal Processing and Machine Learning

Muhammad Omais

*School of Electrical Engineering and Computer Science,
National University of Science and Technology,
Islamabad, Pakistan*
owaseem.bee21seecs@seecs.edu.pk

Muhammad Huzaifa

*School of Electrical Engineering and Computer Science,
National University of Science and Technology,
Islamabad, Pakistan*
mhuzaifa.bee21seecs@seecs.edu.pk

Ahmed Razi Ullah

*School of Electrical Engineering and Computer Science,
National University of Science and Technology,
Islamabad, Pakistan*
aullah.bee21seecs@seecs.edu.pk

Wajiha Ali

*School of Electrical Engineering and Computer Science,
National University of Science and Technology,
Islamabad, Pakistan*
wali.bee21seecs@seecs.edu.pk

Abstract—Induction motors are one of the most crucial electrical equipment and are extensively used in industries in a wide range of applications. This paper presents a machine learning model for the fault detection and classification of induction motor faults by using three phase currents and vibration as inputs. The aim of this work is to protect vital electrical components and to prevent abnormal event progression through early detection and diagnosis. This work presents a machine learning model to detect some of the commonly occurring electrical faults like bearing faults, winding faults and rotor bar faults. The model so developed analyses the voltage and current values given at a particular instant and classifies the data into no fault or the specific fault.

Index Terms—Induction motor, fault analysis, machine learning, predictive maintenance.

I. INTRODUCTION

Induction motors are the most extensively used electrical motors because of their simple construction, ruggedness and low cost. More than 90% of the industries use induction motors, mostly as electrical drives since they can be designed for a wide range of power ratings. In spite of their flexibility and robustness, they are subjected to many catastrophic failures. Identifying these faults at an early stage and attending to them is very important as otherwise they lead to huge production and financial losses. Pre-fault detection and isolation of the healthy parts also prevents fault progression and failure of other more vital components. Industries use a large number of motors and hence, their manual maintenance is tedious and unreliable. Therefore, many attempts have been made towards automatic maintenance. Earlier, conditional monitoring of electrical machines was employed and the electromechanical relays were put to use to achieve this. But these relays are slow in operation and also lead to huge power losses due to mechanical parts involved [6]. Hence, they cannot be used in critical applications requiring small response times.

Electromechanical relays were later replaced by solid state relays as they consume very less power and are comparatively

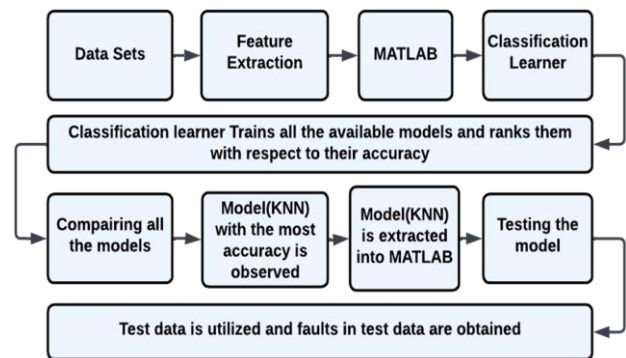


Fig. 1. Block Diagram

fast. With the advent of microprocessors, attempts were made for conditional monitoring of machines by using pre-written programmes downloaded onto the microprocessor chips. The above mentioned techniques, however could not guarantee maximum safety and reliability as they cannot address catastrophic failures. Computer revolution with the dawn of machine learning drew the attention of scientists and they began to think of ways through which these techniques could be used to monitor and safeguard machines. Machine learning models take the place of human to intelligently monitor and maintain the specified system tasks [4]. The aim of this work is to prevent fault progression and protect vital components of the power system by early detection of electrical faults of three phase induction motors using machine learning.

II. BLOCK DIAGRAM

The generic block diagram of the project processes is shown in figure 1.1

III. EXPERIMENTAL SETUP

A. Bearing

Four bearings were installed on a shaft. The rotation speed was kept constant at 2000 RPM by an AC motor coupled to the shaft via rubber belts. A radial load of 6000 lbs is applied onto the shaft and bearing by a spring mechanism. All bearings are force lubricated. PCB 353B33 High Sensitivity Quartz ICP accelerometers were installed on the bearing housing (two accelerometers for each bearing [x- and y-axes] for data set 1, one accelerometer for each bearing for data sets 2 and 3). All failures occurred after exceeding designed life time of the bearing which is more than 100 million revolutions. Three (3) data sets were included. Each data set describes a test-to-failure experiment. Each data set consists of individual files that are 1-second vibration signal snapshots recorded at specific intervals. Each file consists of 20,480 points with the sampling rate set at 20 kHz. The file name indicates when the data was collected. Each record (row) in the data file is a data point. Data collection was facilitated by NI DAQ Card 6062E. Larger intervals of time stamps (showed in file names) indicate resumption of the experiment in the next working day.

B. Winding

The test bench used to acquire the data set was composed of two similar triphasic squirrel cage induction machines, two frequency converters, a failure emulation control panel, and a resistor load bank. One of the induction machines was properly prepared to enable the emulation of stator winding interturn short-circuit. Its stator circuit was re-winded, making it possible to access the ramifications of the winding, in order to insert inter-turn short circuits. Different levels of short-circuit can be emulated, from very incipient defect to severe situation. It operates as a motor and the other machine emulates the mechanical load of the motor. The frequency converters are used to drive the induction machines. This way, the machines can work at different driving frequencies. The induction machines used have the following specifications: 4 poles, 1 HP of mechanical power, delta configuration, 220V of supply voltage, and 3A of rated current. 2240 of fault conditions, distributed into 6 defective classes: high impedance fault of levels 1, 2, and 3; and low impedance fault of levels 1, 2, and 3. For each class, there are patterns acquired with no mechanical load attached, 50% of the rated load, and 100% of the rated load. The driving frequency also varied from 30 Hz to 60 Hz, with steps of 5 Hz. Out of these only 150 were used for more accurate readings.

C. Rotor Bar

The data set contains electrical and mechanical signals from experiments on three-phase induction motors. The experimental tests were carried out for different mechanical loads on the induction motor axis and different severity's of broken bar defects in the motor rotor, including data regarding the rotor without defects. Ten repetitions were performed for each experimental condition. The experimental workbench consists of a three-phase induction motor coupled with a direct-current

	Max	Min	Mean	Std	RMS	Skewness	Kurtosis	Crest Factor	Form Factor	Fault
2004-02-18 00:32:00	0.481	-0.645	-0.002109	0.108896	0.108914	0.043103	1.006777	4.416327	-51.634520	Outer Race
2004-02-18 00:42:00	0.508	-0.537	-0.002089	0.107409	0.107426	0.022695	0.906887	4.728821	-51.417228	Outer Race
2004-02-18 00:52:00	0.554	-0.540	-0.001542	0.108719	0.108727	0.080662	1.060355	5.095329	-70.501831	Outer Race
2004-02-18 01:02:00	0.591	-0.461	-0.002320	0.110181	0.110203	0.047596	1.030171	5.363826	-47.495934	Outer Race
2004-02-18 01:12:00	0.498	-0.630	-0.002595	0.109821	0.109849	-0.018423	0.918053	4.533508	-42.338963	Outer Race
2004-02-19 05:42:00	4.998	-5.000	-0.002752	0.725014	0.725001	-0.510519	12.576183	6.893779	-263.426419	Outer Race
2004-02-19 05:52:00	2.688	-2.881	-0.004487	0.462001	0.462012	-0.325344	3.759312	5.818035	-102.977656	Outer Race
2004-02-19 06:02:00	3.301	-3.696	-0.001703	0.483844	0.483835	-0.377068	4.890965	7.233932	-284.094983	Outer Race
2004-02-19 06:12:00	0.005	0.000	0.001857	0.000987	0.002103	0.579656	3.636864	2.377543	1.132578	Outer Race
2004-02-19 06:22:00	0.002	-0.005	-0.001162	0.001000	0.001533	0.317009	-1.609910	1.304880	-1.319089	Outer Race

180 rows x 10 columns

Fig. 2. Enter Caption

machine, which works as a generator simulating the load torque, connected by a shaft containing a rotary torque wrench. Induction motor: 1hp, 220V/380V, 3.02A/1.75A, 4 poles, 60

Hz, with a nominal torque of 4.1 Nm and a rated speed of 1715 rpm. The rotor is of the squirrel cage type composed of 34 bars. Load torque: is adjusted by varying the field winding voltage of direct current generator. A single-phase voltage variator with a filtered full-bridge rectifier is used for the purpose. An induction motor was tested under 12.5, 25, 37.5, 50, 62.5, 75, 87.5 and 100% of full load. Broken rotor bar: to simulate the failure on the three-phase induction motor's rotor, it was necessary to drill the rotor. The rupture rotor bars are generally adjacent to the first rotor bar, 4 rotors have been tested, the first with a break bar, the second with two adjacent broken bars, and so on rotor containing four bars adjacent broken.

IV. PROCEDURE

A. Bearing

The data was collected from each file and features were extracted then according to those feature healthy and unhealthy faults were separated. These were then fed into the classification learner on Matlab which trained multiple models and chose the one with the highest accuracy

1) Data Generation and Preprocessing :

The data set is generated using a three-phase induction motor. Different health conditions are simulated and 4 tests are conducted on each of the 4 bearings. The 4 faults that are to be tested are Inner race fault, outer race fault on which 2 tests are conducted, and the Roller element fault. All these faults are compared with the normal bearing data.

2) Feature Extraction :

Different Features were extracted from the given data set of each bearing which will contain data for bearing 1, bearing 2, bearing 3 and bearing 4. Features like minimum value, maximum value, Standard deviation, skewness, crest factor, form factor etc are extracted and can be compared to identify which bearing will have the maximum deviation from the normal data set value and hence we can confirm that the bearing with the maximum deviation will have the fault in it which will be identified using our machine learning model.

3) Classification Learner App Integration :

After the features are extracted and a .csv file is made for processing it is imported into the MATLAB workspace and hence then is fed to classification learner app. Exporting Features to

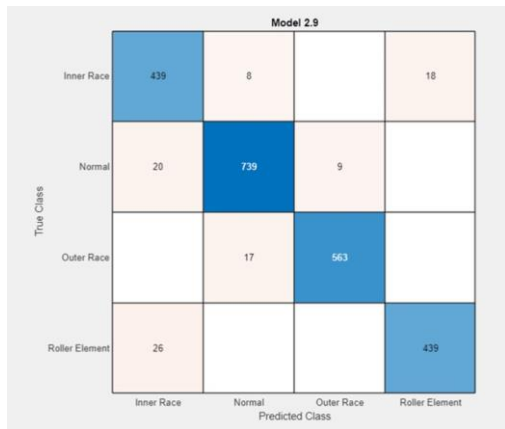


Fig. 3. fig 2

Classification Learner App : The top features are exported to the Classification Learner app to build a predictive model for fault classification. The features are used to train a decision tree model, and the accuracy of the model is evaluated. Each model has its own accuracy based upon its operation and data set evaluation, and the best among them is to be selected, KNN in the case of Bearing faults. The KNN model has 95.7% accuracy, indicating a higher accurate response for larger data set. The confusion matrix of the KNN is observed and can be seen in fig2. The model correctly calculated 439 faults as inner faults since they were inner faults, and miscalculated 8 inner faults as normal bearing data, 18 inner race faults as roller element fault, Similarly, the model correctly calculated 563 outer race faults as outer race faults, and miscalculated 7 Outer race faults as Normal bearing data. In doing so we have successfully trained our model which displays a much higher accuracy.

Exporting the KNN model to MATLAB : After training the model, the KNN model is exported to MATLAB workspace and

ultimately its accuracy is validated using test data. The data set initially set apart from the training data, is used in the same manner training data was used. Features are extracted from the testing data, and a certain csv file is generated with 9 features, parameters of testing data and training data are same. The csv file generated is then fed to the model and then ultimately the model will test on the provided test data and display the fault in any one of the bearing whose features are extracted from the test data in the first place meaning if the feature extraction is done for bearing 3, the model will test and identify the faults, if any in this bearing. If there are no faults in a bearing it will label it as normal

B. Winding

The data was collected from each file and the parameters were fed into a diagnostic feature designer app on Matlab which extracted and ordered the most useful features from it. From there we send the most important features and send them to the classification learner to get the model of the highest accuracy.

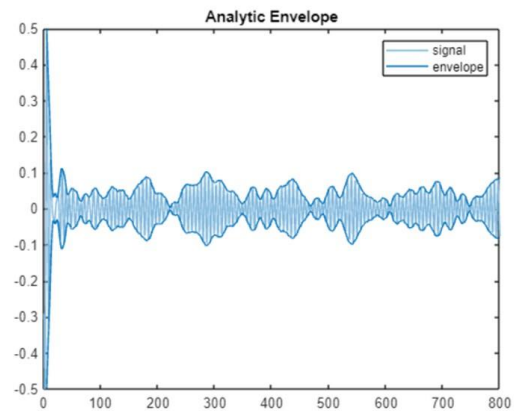


Fig. 4. Enter Caption

C. Broken Bar

The useful data was separated from the data set . Performed wavelet transformation on current and vibration signals and the area of maximum power was extracted. The extracted spectrums were sent to the diagnostic feature designer app and feature extraction was performed and sent to classification learner like we did in the winding.

1) Data Generation and Preprocessing :

Data Set Information The data set is generated using a three-phase induction motor coupled with a direct-current machine, simulating load torque. Different health conditions are simulated by introducing 0 to 4 broken rotor bars. Signals include voltage of each phase, current draw of each phase, and various mechanical vibration speeds under different health and load conditions. **Ensemble Member Data Generation :** The

script extracts relevant information from the provided data files, associates them with the number of broken rotor bars (health condition), and generates individual ensemble member data files for each combination of health, load torque, and experiment index. The data files are saved in a target folder for further processing. File Ensemble Datastore :The file ensemble datastore is created to efficiently handle and process the ensemble member data files. It includes functions for reading and writing data .

- 2) *Diagnostic Feature Designer App Integration* : Importing Data into Diagnostic Feature Designer App : The ensemble data is imported into the Diagnostic Feature Designer app for further analysis. Time-domain signals, envelope signals, and envelope spectrum data are selected and configured for processing .Creating Predictive Features :The script demonstrates how to create predictive features using the Auto Features option for time-domain signals and a manual approach for envelope spectrum data. Features are generated based on One-way ANOVA ranking to identify the most significant features for fault classification.

3) Classification Learner App Integration :

The top features are exported to the Classification Learner app to build a predictive model for fault classification. The features are used to train a decision tree model, and the accuracy of the model is evaluated.

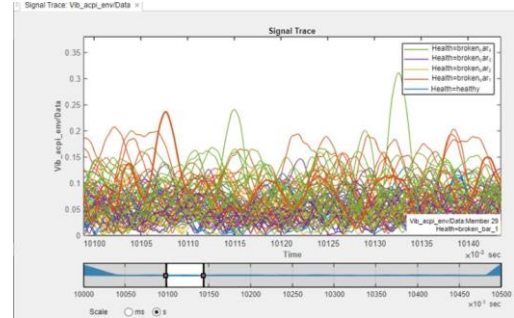


Fig. 5. Enter Caption

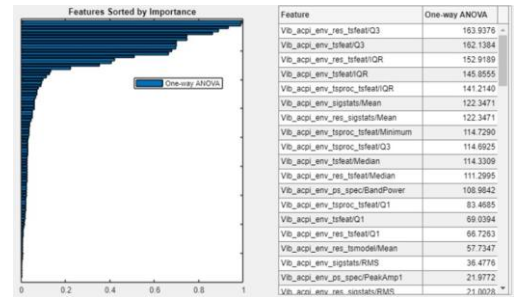


Fig. 6. Enter Caption

V. CONCLUSION

A. Bearing

After feature extraction it was clearly visible that when the faults have occurred. As a result when the faults were classified and put into classification learner an accuracy of 95.2% was achieved.

B. Winding

When the features of the data were put into classification learner an accuracy of 95% was achieved which was later tested onto a data an 1 out of 27 was falsely classified into a wrong class.

C. Broken Bar

The predictive maintenance approach aims to classify the health condition of the induction motor based on features

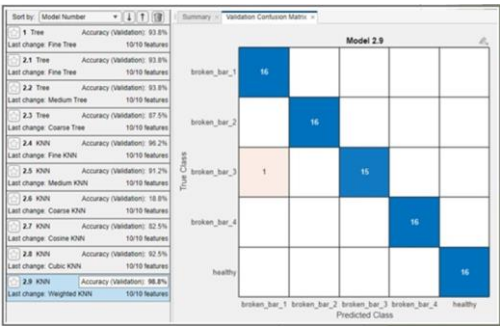


Fig. 7. Enter Caption

extracted from electrical and vibration signals. When the features of the data were put into a classification learner an accuracy of 96% was achieved.

REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.
- [2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] I. S. Jacobs and C. P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
- [4] Y. Lei, F. Jia, X. Zhou, and J. Lin, "A Deep Learning-based Method for Machinery Health Monitoring with Big Data", Journal of Mechanical Engineering, vol. 51, pp. 49-56, Nov. 2015.
- [5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.
- [6] Faiz, J., Ghorbanian, V., Ebrahimi, B. "A survey on condition monitoring and fault diagnosis in line-start and inverter-fed broken bar induction motors". IEEE Int. Conf. on Power Electronics, Drives and Energy Systems (PEDES), 2012, pp. 15
- [7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.